# MPG No. 6.3.19 2′2′2 (2’2’2 setting) [ Type III, orthorhombic ] [G tensor]

* Rank 0 tensor.

$$[G]$$

$$G = \mathbb{G}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} G_{xx} & 0 & 0 \\ 0 & G_{yy} & 0 \\ 0 & 0 & G_{zz} \end{bmatrix}$$

$$G_{xx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$
$$G_{yy} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$
$$G_{zz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{yzx} \end{bmatrix}$$

$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{yzx} = 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$G_{zxy} = -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$
$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$
$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & 0 & 0 \\ G_{xxyy} & G_{yyyy} & G_{yyzz} & 0 & 0 & 0 \\ G_{xxzz} & G_{yyzz} & G_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{xxxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$
$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)}$$
$$G_{xxzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$
$$G_{yyyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$
$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$
$$G_{zzzz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$
$$G_{yzyz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - \mathbb{G}_v^{(2)}$$
$$G_{zxzx} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + \mathbb{G}_v^{(2)}$$
$$G_{xyxy} = \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & G_{xxyy} & G_{xxzz} & 0 & 0 & 0 \\ -G_{xxyy} & 0 & G_{yyzz} & 0 & 0 & 0 \\ -G_{xxzz} & -G_{yyzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxyy} = -2\mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$
$$G_{xxzz} = 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$
$$G_{yyzz} = 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} G_{yzyz} & 0 & 0 \\ 0 & G_{zxzx} & 0 \\ 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{yzyz} = \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(4)}$$
$$G_{zxzx} = \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 2\mathbb{G}_v^{(4)}$$
$$G_{xyxy} = \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzyz} & 0 & 0 \\ 0 & G_{zxzx} & 0 \\ 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{yzyz} = -3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$
$$G_{zxzx} = 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$
$$G_{xyxy} = 2\mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & G_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{yzyz} = -3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{zxzx} = 3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{xyxy} = 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ G_{yyxx} & G_{yyyy} & G_{yyzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ G_{zzxx} & G_{zzyy} & G_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{yzyz} & 0 & 0 & G_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{zxzx} & 0 & 0 & G_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{xyxy} & 0 & 0 & G_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)} \\
G_{xxyy} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)} \\
G_{xxzz} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)} \\
G_{yyxx} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} + 2\mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)} \\
G_{yyyy} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)} \\
G_{yyzz} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)} \\
G_{zzxx} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)} \\
G_{zzyy} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)} \\
G_{zzzz} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)} \\
G_{yzyz} &= \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)} \\
G_{yzzy} &= \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)} \\
G_{zxzx} &= \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)} \\
G_{zxxz} &= \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)} \\
G_{xyxy} &= \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)} \\
G_{xyyx} &= \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & G_{yzyz} & 0 & 0 & G_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{zxzx} & 0 & 0 & G_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{xyxy} & 0 & 0 & G_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
G_{yzyz} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{yzzy} &= -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{zxzx} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{zxxz} &= -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{xyxy} &= \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{xyyx} &= -\mathbb{G}_0^{(3)} - 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} G_{xxxx} & 0 & 0 \\ 0 & G_{yyyy} & 0 \\ 0 & 0 & G_{zzzz} \\ 0 & 0 & G_{yyzz} \\ G_{zzxx} & 0 & 0 \\ 0 & G_{xxyy} & 0 \\ 0 & G_{yzzy} & 0 \\ 0 & 0 & G_{zxxz} \\ G_{xxyy} + G_{yyzz} - G_{yzzy} - G_{zxxz} + G_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxx} = 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$

$$G_{yyyy} = 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$

$$G_{zzzz} = 3\mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$

$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)}$$

$$G_{zzxx} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)}$$

$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)}$$

$$G_{yzzy} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)}$$

$$G_{zxxz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)}$$